Fiscal Year:	FY 2018	**Task Last Updated:**	FY 05/01/2018
PI Name:	Hammond, Timothy M.D.		
Project Title:	Evolution of Genotypic and Phenotypic Changes in Yeast Related to Selective Growth Pressures Unique to Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/17/2012	**End Date:**	07/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 7/31/2018 per PI (Ed., 7/17/17); NOTE extension was to 7/15/2018 per NSSC information (Ed., 4/24/18) NOTE: End date is now 7/15/2017, per NSSC information (Ed., 7/18/16) NOTE: End date is now 7/15/2016, per NSSC information (Ed., 4/20/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Birdsall, Holly M.D., Ph.D. (Baylor College of Medicine) Nislow, Corey Ph.D. (Stanford University/University of British Columbia) Giaever, Guri Ph.D. (University of British Columbia)		
Grant/Contract No.:	NNX12AM93G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

This application investigates microbial growth and physiological responses to the multiple stimuli encountered in space flight environments, using Saccharomyces cerevisiae as a model system. Yeast are an ideal model because their complete gene sequence is known; there are extensive and sophisticated clone libraries available for study; and they can easily be flown in space, held at +4oC in a quiescent state, and activated by simple warming. Regulatory mechanisms in yeast are highly similar to mammalian cells. We have shown that microgravity activates a unique set of genes in yeast, but the specific molecular mechanisms remain incompletely defined.
In a corollary set of experiments, we are utilizing next generation sequencing techniques to assess the effects of long term exposure to radiation on the yeast model organism Saccharomyces cerevisiae. The cellular response to DNA damage has been studied extensively in a variety of model systems. The most comprehensive view of this process has come from yeast which, owing to the conservation of this essential pathway, shares homologs of their human DNA repair counterparts. A key observation from our group and others is that different types of DNA damage require unique repair genes and proteins. Accordingly, we are studying the cellular response to energetic particle radiation experienced in space using whole genome sequencing of thousands of yeast cells exposed to a large matrix of space-specific conditions.

We hypothesize that by identifying specific regulatory motifs within the promoter regions of microgravity-responsive genes, we will be able to determine transcription factors that bind the motifs, and thus the signaling pathways involved in microbial growth and physiological responses. We propose ground-based studies with transition to a future flight experiment, designed so as to have a high probability of obtaining genome wide data.

The proposed experiments are timely because (a) recent improvements in promoter analysis have vastly increased the accuracy of promoter elements predictions; (b) novel yeast clones allow direct measurement of gene expression by fluorescence using the plate reader in the Advanced Biological Research System (ABRS) that is already in place on the International Space Station (ISS); (c) yeast on agar plates can be assayed repetitively, thereby eliminating sample to sample variability to increasing the statistical power of the data; (d) new reagents allow design of experiments with minimal up mass (less than 2 kg); and (e) all data can be returned by telemetry, eliminating the need for down mass.

To test our hypothesis, we will:

1) Identify promoter motifs associated with genes responding to Rotating Wall Vessel (RWV)-emulated microgravity compared to static cultures. The RWV will be used to emulate microgravity as we have already demonstrated that it induces changes in yeast gene expression that are similar to those of yeast flown in space. The Yeast Green Fluorescent Protein (GFP) clone collection will permit identification, by changes in fluorescence, of genes whose expression changes in response to real and simulated microgravity. After extensive normalization and analytic procedures, microgravity-responsive genes will be grouped into clusters on the basis of similar kinetics, under the assumption that these are likely to be co-regulated. We will then search the upstream sequence of co-expressed genes and identify candidate binding motifs for transcription factors.

2) Confirm the role of regulatory sequences in directly activating transcription of a set of genes changed by RWV-simulated microgravity and to identify transcription factors that interact with these regulatory motifs. This specific aim makes novel use of the yeast deletion collection, a systematic gene-by-gene set of complete gene deletions, the only such resource that exists for any group of organisms. Transcription factors will be identified by robotically introducing a reporter plasmid that codes for 1) a suicide-inducing enzyme that is activated in all clones except the one(s) with a deletion in the gene for the transcription activating factor that binds to the promoter motif or 2) an endogenously activated essential gene to identify repressors.

3) Confirm and extend the results obtained in our ground-based studies on the selective growth pressure in the true microgravity environment. Utilizing the genome wide yeast GFP clone collections spotted on agar plates, we propose a parallel flight study. Our question can ultimately only be definitively answered in the true microgravity of space, because on the ground gravity must always be onset with an equal and opposite force.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This work aims to understand the genetic control mechanism that controls the gene expression response to cellular responses to microgravity in yeast.
Yeast have 70% homolog with the human genome, and are a useful model system to define primary and secondary drug metabolism pathways that cannot be done in current mammalian cell culture systems.

The short term aim is to map the genetic promoter elements that mediate gene responses to microgravity. The long term goal is to use this information to unveil drug metabolism pathways in the unique metabolic environment provided by microgravity.

We propose to establish a sustained genomics platform on the ISS to understand the long term effects of the radiation, and other features of the ISS environment, on crew health by dissecting the essential processes of DNA damage and DNA repair. This has broad Earth benefits in understanding effects and optimization of medically necessary radiation therapy and chemotherapy with alkylating agents.

The APEX-02-2 work quantitatively measures radiation damage to yeast DNA exposed to space radiation. Using state of the art technologies, the science team can, for the first time, conduct a highly powered, genome wide analysis of mechanisms of radiation damage in space. Knowledge gained from this investigation will build upon previous data to understanding the mechanisms of radiation damage, and may provide simple approaches to ameliorating space based radiation damage.

By using yeast strains to understand the effects of space based radiation stresses at the cellular level, a greater knowledge of the regulatory mechanisms at work in cells will be gained. A hallmark of radiation stresses is that they induce physiological responses; primary among them are DNA damage. As yeast is a model eukaryotic organism, DNA damage changes caused by radiation could lead to a greater understanding of these processes on Earth, thus benefiting all citizens. Knowledge gained from this investigation will build upon previous data to understanding the mechanisms of

	radiation damage, and may provide simple approaches to ameliorating clinical radiation damage. Yeast deletion project is a project to create data for a near-complete collection of gene-deletion mutants of the yeast Saccharomyces cerevisiae. Each strain carries a precise deletion of one of the genes in the genome, and has been functionally characterized. We have now cloned the 6,000+ strains of the yeast deletion series into the background of 10 separate deletion clones of genes known to play a major role in responses to radiation damage.
Task Progress:	Results and discussion of work completed. Preflight studies validated assays for reactive oxygen species and glutathione in Saccharomyces cerevisiae during microgravity simulation; established the timing, optimal media (rich versus poor) and dynamic range of rotational interventions for low redox potential in giant yeast colonies, and determined if lack of convection of gaseous ammonia in true microgravity changes the stratification and expansion of in giant yeast models of differentiation and tumorigenesis. Fight definition, flight activities, ground control, and sample recovery are complete. There were zero defects in the SpaceX-11 flight operations and experiment execution. Sample processing by DNA extraction, a PCR amplification of reporter barcodes, next generation sequencing, and gene set enrichment analysis are complete. The final Cytoscope/Cluego visualization and probability modeling are ongoing. The resources provided have allowed us to address the unifying hypothesis in a sophisticated and nuanced manner, with some important caveats for countermeasures. Our hypothesis is that cells cope with the stress associated with the microgravity of space environment by selectively changing the expression of specific target genes and this response is mediated by defined transcription factors. We have been able to determine specific target genes and mechanistic pathways determining them, including transcription factors, in a variety of space based models using background deletion of several radiation protection genes, as well as stimulation of the system with multiple drugs. In particular, we can now address whether metformin, a drug originally utilized in diabetes, but shown to have cancer protective properties, may be a useful radiation countermeasure. In this manner, the work directly addresses multiple Space Biology Science Plan's research goals, objectives, and guiding questions. All OMICS data will be annotated and deposited in the NASA GeneLab portal.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
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